

Proofs by Structural Induction

```
theory Structural_Induction
  imports Main
begin
```

Defining a well-formed (recursive) algebraic data type automatically, in Isabelle/HOL, leads to the generation of an induction principle. For instance, for the following binary trees data type

```
datatype 'a tree = Leaf | Node "'a tree" 'a "'a tree"
```

that induction principle is

$$\frac{P \text{ Leaf}; \quad [\forall t_1 t_2 a. P t_1; P t_2 \implies P \text{ Node } t_1 a t_2]}{P t}.$$

As a general rule of thumb, you should follow the same heuristics for induction: manipulate the goal until you can apply the Induction Hypothesis (I.H.). In this example, you have two induction hypotheses, one for each subtree.

The following is an example proof by structural induction on trees. After replicating it on your own, take a close look at the two goals: the base case and the step case, and note the two I.H.'s, each corresponding to a subtree.

Exercise: write down a detailed manual Isabelle/HOL proof using only forward reasoning (i.e. `OF`), backward reasoning (i.e. `rule`), and substitution (i.e. `subst`).

```
lemma "set (inorder t) = tree_set t"
proof(induction t)
  case Leaf
  then show ?case
    by auto
next
  case (Node t1 x2 t2)
  then show ?case
    by auto
qed
```

```
end
```